

List of Steps

1. Install OSS CAD Suite and add to path.
2. Create your own directory for the particular device.
3. Write the behavioral source code file and save it as device.v
4. Write the testbench source code file and save it as device_tb.v
5. Open command prompt and call the environment.bat file as illustrated.
6. Write `iverilog -o simulation.vvp device.v device_tb.v`
7. Write `vvp simulation.vvp`
8. Write `gtkwave simulation.vcd` and see the waveform.
9. Write `yosys` to enter yosys mode in the terminal.
10. Write `read_verilog device.v`
11. Write `hierarchy -top device.v`
12. Write `proc opt techmap opt`
13. Write `write_verilog device_netlist.v`
14. Write `show`
15. Write `exit` to jump out of yosys mode.
16. Write `dot -Tpng show.dot -o schematic.png`
 - a. or add dot folder address to PATH if error occurs.
 - b. or copy the address of dot as path, then write as follows in the terminal: `"...\graphviz\Graphviz-12.0.0-win64\bin\dot.exe" -Tpng show.dot -o schematic.png`

OR Gate

Truth table

OR Gate		
a	b	y
0	0	0
0	1	1
1	0	1
1	1	1

Behavioral source code (or_gate.v)

```
module or_gate(y,a,b);
output y;
input a,b;
assign y=a|b;
endmodule
```

Testbench source code (or_gate_tb.v)

```
`timescale 1ns/1ps

module or_gate_tb;
    // 1. Creating registers for inputs and a wire for the output
    reg test_a;
    reg test_b;
    wire test_y;

    // 2. Instantiating the actual OR gate module (connect pins)
    or_gate uut (
        .y(test_y),
        .a(test_a),
        .b(test_b)
    );
```

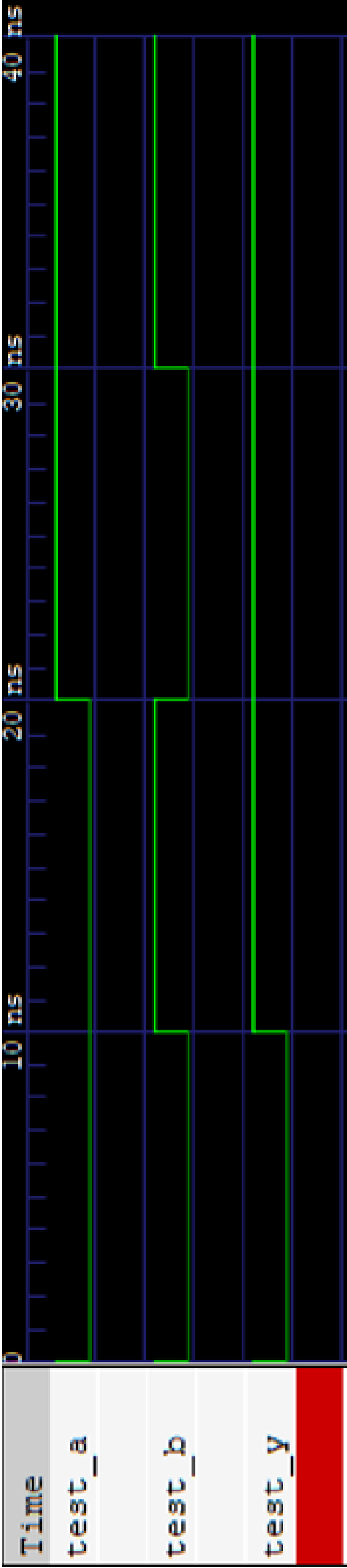


Fig. 1(a) - OR gate output waveform

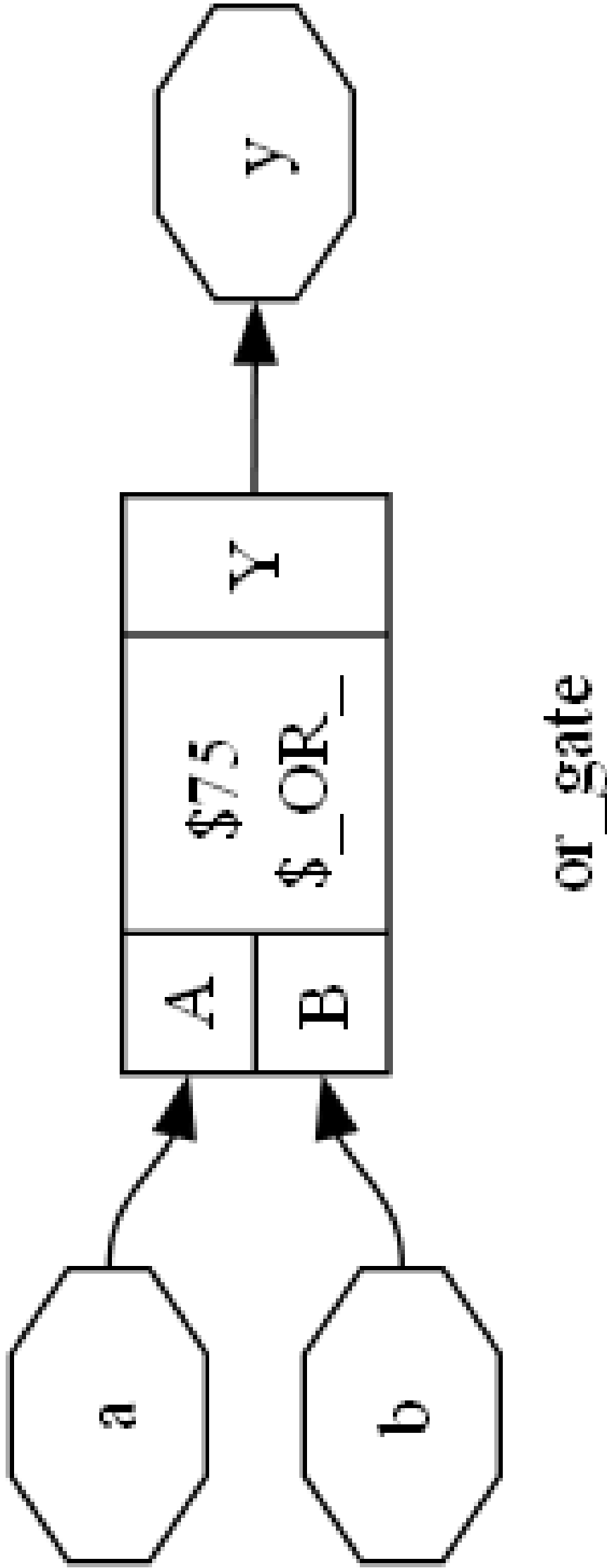


Fig. 1(b) - OR gate generated schematic

```
// 3. Generate the input signals over time
initial begin
    // Telling iverilog to dump waveform data for GTKWave
    $dumpfile("simulation.vcd");
    $dumpvars(0, or_gate_tb);

    // Test Case 1: Both 0
    test_a = 0; test_b = 0; #10;

    // Test Case 2: 0 and 1
    test_a = 0; test_b = 1; #10;

    // Test Case 3: 1 and 0
    test_a = 1; test_b = 0; #10;

    // Test Case 4: Both 1
    test_a = 1; test_b = 1; #10;

    $finish; // End simulation
end
endmodule
```

fgfg

```
call C:\oss-cad-suite\environment.bat
```

fgfg

```
iverilog -o simulation.vvp or_gate.v or_gate_tb.v
```

fgfg

```
vvp simulation.vvp
```

Generate & check output waveform - Fig. 1(a)

```
gtkwave simulation.vcd
```

```
yosys
```

```
/-----\
| yosys -- Yosys Open SYnthesis Suite          |
| Copyright (C) 2012 - 2026  Claire Xenia Wolf <claire@yosyshq.com> |
| Distributed under an ISC-like license, type "license" to see terms |
\-----/
Yosys 0.68+40 (git sha1 0f2bcb94b-dirty, Release, GNU /usr/bin/x86_64-w64-mingw32-g++ 15.2.1)
```

```
yosys> read_verilog or_gate.v
```

```
yosys> hierarchy -top or_gate
```

```
yosys> proc opt techmap opt
```

```
yosys> write_verilog or_gate_netlist.v
```

```
yosys> show
```

```
5. Generating Graphviz representation of design.
Writing dot description to 'show.dot'.
Dumping module device to page 1.
```

```
yosys> exit
```

```
Warnings: 2 unique messages, 3 total
End of script. Logfile hash: a730de9a18
Yosys 0.68+40 (git sha1 0f2bcb94b-dirty, Release, GNU /usr/bin/x86_64-w64-mingw32-g++ 15.2.1)
Time spent: 5% 2x write_verilog (0 sec), 5% 2x read_verilog (0 sec), ...
```

```
[OSS CAD Suite] D:\[redacted] \AND>"D:\[redacted]
[redacted]\windows_10_cmake_Release_Graphviz-15.1.1-win64\Graphviz-
15.1.1-win64\bin\dot.exe" -Tpng show.dot -o schematic.png
```

Generate & check or_gate schematic - Fig. 1(b)

